

Electric Airbike

Engineering Design & Analysis — Full Technical Deep Dive Example

Project statement: Given fixed starting positions for the wheels, control device (handlebars and connected rod), and motor, design an electric airbike incorporating the required vehicle systems and evaluate the resulting designs through CAD, structural, fluid, and thermal studies.

Project	Electric Airbike Design
Primary CAD	SolidWorks
Material studies	6061-T6 / 7075-T6 aluminum; other materials as modeled
Analysis	FEA, flow simulation, nonlinear study, fatigue study, thermal study
My role	Design, CAD integration, structural analysis, thermal/flow studies

This example intentionally reorganizes the original 136-slide presentation into a portfolio case study. The goal is not to reproduce the presentation; it is to show how the same engineering work can be presented as a coherent design story.

1. Requirements & Design Constraints

The project began with fixed locations for the wheels, handlebars/control linkage, and motor. The chassis therefore had to be designed around existing geometric constraints while preserving operator space and room for future components. The presentation also identifies the chassis material as Aluminum 6061-T6, with 2.5 in OD $\times$ 0.25 in wall round tube for the main chassis and 2.0 in OD $\times$ 0.25 in wall tube for the later rear supports. ■

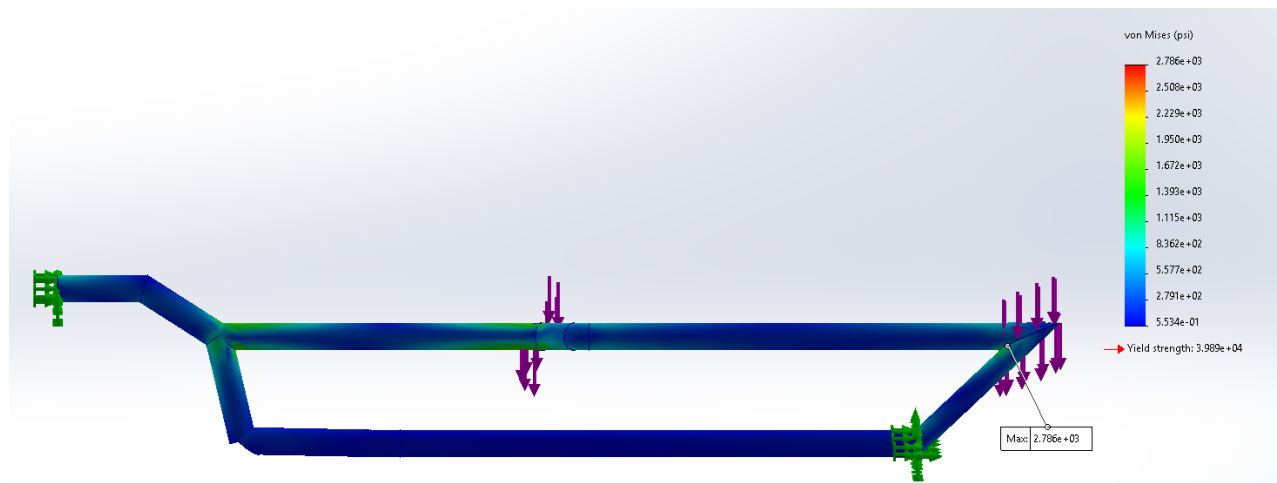
Design constraints carried through the project:

- Fixed wheel and control-device locations
- Motor mounting/interface requirements
- Operator packaging and foot clearance
- Space for future components and seat hardware
- Structural strength and acceptable deflection
- Manufacturable geometry and practical material selection

The original presentation notes that the rear wheel clearance was relatively tight and identifies this as an area that would need more tolerance in a real production design.

2. Chassis Design & Structural Iteration

The first major design iteration was evaluated without the rear vertical supports. The analysis showed that the initial chassis was substantially under-supported, so two rear supports were introduced and the model was re-evaluated. This is the clearest example in the project of the design-analysis-redesign loop that should be emphasized in the portfolio.



Structural results: maximum stress decreased from 5.233×10^4 psi to 2.786×10^3 psi after the support modification. Maximum deflection decreased from 5.512 in to 0.03444 in. Minimum factor of safety increased from 0.76 to 14.

The presentation attributes the loading to a 200 lb operator and a 400 lb fan assembly. The final design therefore demonstrates how simulation was used to identify a major structural weakness and guide a specific geometric modification rather than simply documenting a final FEA result.

3. Seat Frame Design & Nonlinear Loading

The seat was developed around operator packaging, seating dimensions, clearance from the chassis, and room for cushioning. The modeled seat uses Aluminum 7075-T6 with a 0.2294 in sheet thickness. The initial dimensions were established from an estimated 12 in × 12 in seating area and a 16 in overall height.

The analysis modeled a changing load over a two-second interval. The presentation describes the main seat load decreasing from 200 lb to 150 lb while 50 lb was transferred to the backrest. The initial study produced a factor of safety of 1.862 and identified the need for additional support/suspension to mitigate displacement.

This section is worth keeping in the deep dive because it demonstrates that the seat was not treated as an isolated CAD component: its geometry, operator interface, loading, and later suspension system were developed as connected design problems.

4. Seat Suspension: Iteration & Fatigue Study

The suspension design was integrated into the existing seat/chassis geometry and evaluated under the defined loading and fixture conditions. The second iteration modified the connection geometry after SolidWorks limitations prevented the intended curved connection from being modeled as planned.

Iteration results: maximum stress decreased from 3.013×10^4 psi to 1.788×10^4 psi; maximum displacement decreased from 0.66 in to 0.40 in; and minimum factor of safety increased from 2.432 to 4.097.

An important limitation should remain visible in the portfolio: the presentation reports that the damage and cycle-life plots contained an unresolved error, so fatigue life could not be quantified reliably. The second design nevertheless showed improved stress, displacement, and minimum FOS.

5. Battery Pack & Thermal Study

The battery pack was designed as the vehicle's power source and evaluated with a preliminary thermal model. The study used a convection coefficient of 10 W/m·K and included rudimentary fins as a temporary cooling approach.

The simulated maximum temperature reached 770,100 °F. The presentation explicitly recognizes this as nonphysical for the intended battery system and concludes that the battery would experience catastrophic thermal runaway long before this temperature. The result is therefore best presented as a **modeling limitation / boundary-condition problem**, not as a physical prediction.

For the portfolio, I would keep this result because it demonstrates engineering judgment: the important outcome is not the extreme number itself, but recognizing that the simulation setup does not represent a physically valid operating condition and identifying the need for a new thermal solution.

6. Fixed-Wing Aerodynamics & Structural Analysis

A fixed wing was designed from scratch for the airbike and evaluated using flow simulation. The wing was modeled without onboard flight-control surfaces. The presentation reports that the original submission used 55 mph while the updated study shown in the final presentation used 75 mph.

Reported results: maximum stress of approximately 4,795 psi, maximum displacement of approximately 0.6859 in, and minimum factor of safety of approximately 0.605 for the modeled balsa configuration.

The low FOS led to a material/design change: the wing material was changed to Aluminum 6061-T6 in the larger vehicle model. This also exposed a major vehicle-level optimization problem because the modeled wings were still solid rather than internally ribbed/sparred.

7. Propeller / Fan Flow Study

The propeller/fan system was designed to propel the airbike and included both the blade airfoils and protective cage. A rotating-region flow study was used to examine pressure, force, trajectories, and particle behavior.

The study reported an average normal force of approximately 798 lbf. The presentation then compared this with an overall vehicle mass of approximately 3,000 lb and concluded that the modeled thrust would not be sufficient to move the vehicle as configured.

Rather than treating that as a final failure, the presentation identifies several optimization paths: reducing wing mass, increasing fan size, optimizing blade count/geometry, and selecting lighter materials while maintaining a suitable factor of safety. This is a strong vehicle-level engineering discussion and should remain in the deep dive.

8. Battery-Pack Heat Exchanger

The heat exchanger was developed to manage the battery pack's thermal profile. The flow study modeled R-134a and oil and used mass-flow boundary conditions. The presentation calculated a refrigerant mass flow of 1.17 lb/s and an oil mass flow of 0.35 lb/s for the modeled conditions.

The study produced an R-134a outlet temperature of -9.81 °F. The presentation investigates this result and attributes the localized cooling to pressure-drop/expansion effects associated with the unpressurized outlet boundary condition. It then identifies a more realistic pressurized outlet condition of approximately 50–60 psia as a way to remove that modeling artifact.

9. Final Design & Engineering Assessment

The final portfolio story should not claim that every subsystem is production-ready. The presentation itself identifies several areas requiring further engineering: additional rear-wheel clearance, a more cost-effective structural material selection, improved wing structural architecture and weight reduction, a new battery cooling approach, and further fan/vehicle optimization.

The stronger story is that the project progressed from a set of fixed geometric constraints into an integrated airbike concept, while simulation repeatedly exposed design weaknesses and informed subsequent changes.

Key demonstrated engineering work

Area	Evidence from the project
Chassis	Iterative FEA; support redesign; stress, deflection and FOS comparison
Seat	Operator packaging; nonlinear load distribution; structural analysis
Suspension	Integrated redesign; stress/deflection/FOS improvement; fatigue-study limitations identified
Aerodynamics	Fixed-wing design; flow trajectory/particle studies; structural FEA
Propulsion	Fan/blade design; rotating-region flow study; force assessment
Thermal	Battery thermal study; heat-exchanger flow/thermal analysis

10. My Contribution

Primary responsibilities: chassis CAD and structural iteration; seat and suspension design; integration of vehicle subsystems; setup and interpretation of multiple SolidWorks simulation studies; wing and fan modeling; battery and heat-exchanger studies.

Engineering judgment demonstrated: identifying excessive chassis deflection and adding supports; recognizing the limitations of the fatigue and thermal models; identifying the low wing FOS; recognizing that vehicle mass made the modeled fan force insufficient; and proposing specific redesign directions.

11. Engineering Takeaways

- **Simulation should drive design decisions:** the chassis support redesign is the clearest example of analysis directly changing geometry.
- **Subsystem results must be considered at vehicle level:** the fan study became meaningful only when compared against the approximately 3,000 lb vehicle mass.
- **Model limitations matter:** the thermal and fatigue studies contain explicitly acknowledged limitations, which should be presented rather than hidden.
- **Weight and manufacturability remain future optimization targets:** the final concept provides a basis for further structural and aerodynamic refinement.

12. Supporting Documents

Recommended links on the website: Final presentation PDF • CAD assembly / interactive 3D model • Chassis drawings • Seat/suspension drawings • Simulation result sets • Heat-exchanger study • Any MATLAB/calculation files used in sizing or analysis.

Portfolio note: this deep dive is intentionally shorter and more analytical than the original presentation. The original deck contains the detailed plots, setup screens, meshes, animations, and intermediate results; the website should link those artifacts for readers who want to inspect the evidence.